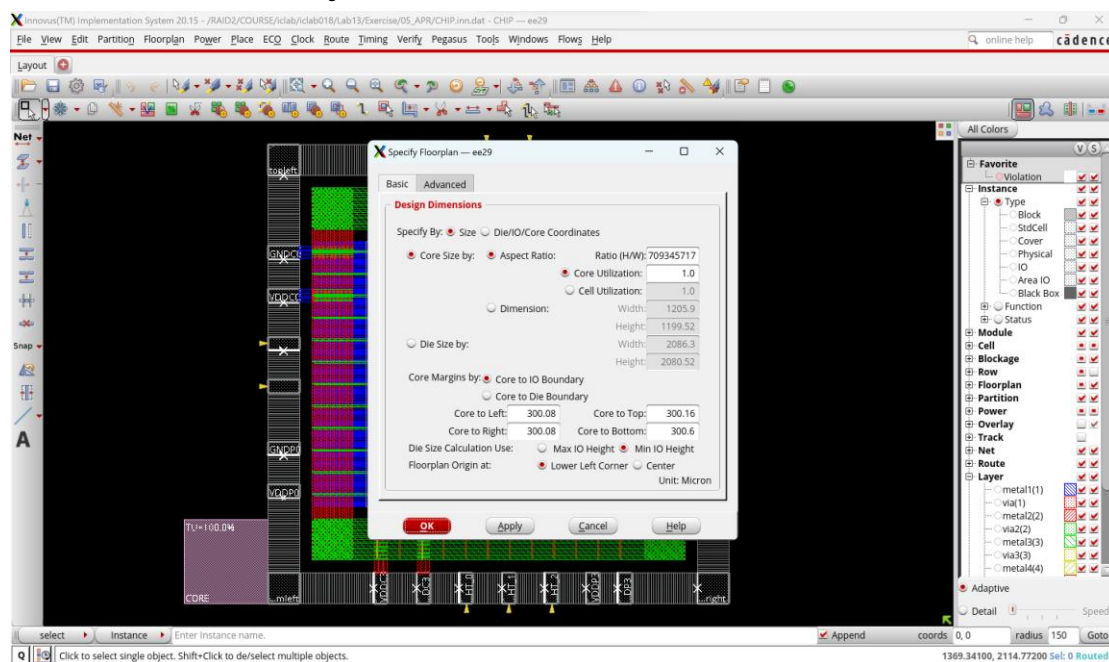
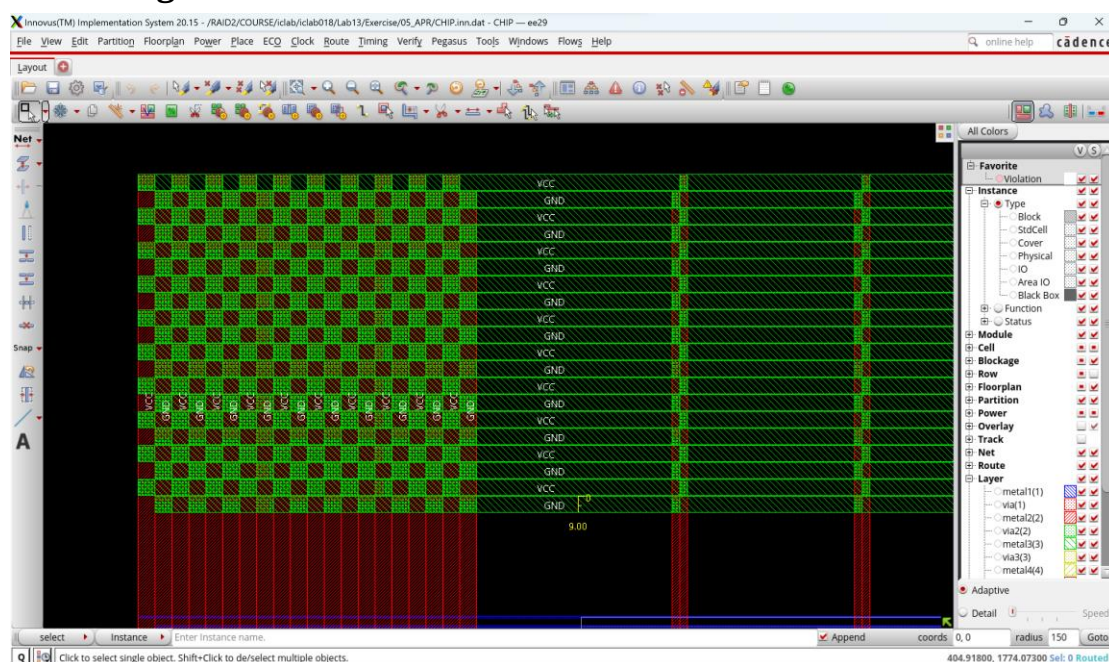


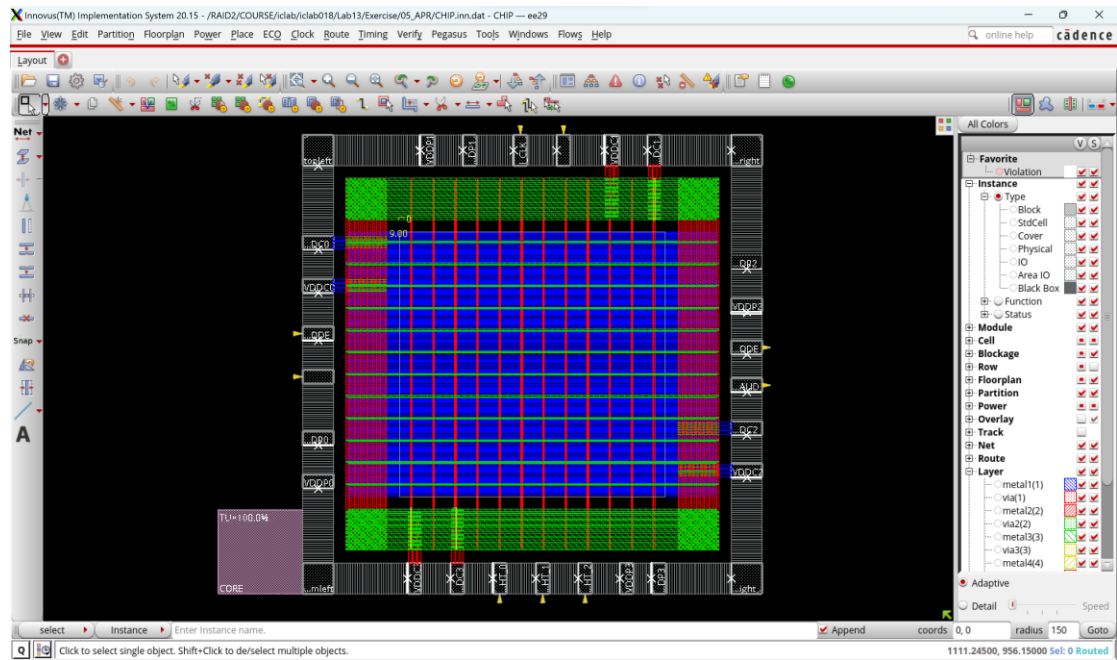
Report

1. Core to IO boundary :



2. Core Ring :





3. Post-Route setup time analysis :

iclab018

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...

iclab018

Glitch Analysis: View av_func mode max -- Total Number of Nets Analyzed = 8904.
Total number of fetched objects 8904
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 8883, 0.1 percent of the nets selected for SI analysis
End delay calculation. (MEM=2339.57 CPU=0:00:00.0 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=2339.57 CPU=0:00:00.1 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:03.9 real=0:00:04.0 totSessionCpu=0:00:47.1 mem=2339.6M)

timeDesign Summary

Setup views included:
av_func_mode_max

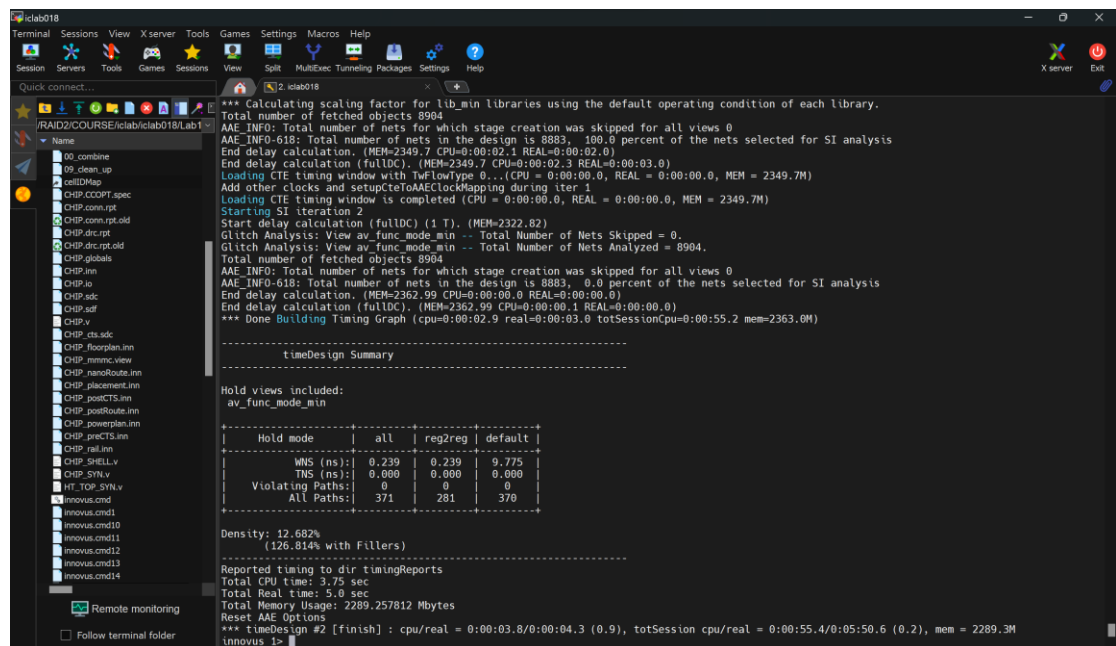
Setup mode	all	reg2reg	default
MNS (ns):	0.730	0.730	1.066
TNS (ns):	0.000	0.000	0.000
Violating Paths:	0	0	0
All Paths:	371	281	370

DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000	0 (0)
max_tfan	0 (0)	0.000	0 (0)
max_fanout	0 (0)	0	0 (0)
max_length	0 (0)	0	0 (0)

Density: 12.682%
(126.814% with Fillers)
Total number of glitch violations: 0

Reported timing to dir timingReports
Total CPU time: 21.42 sec
Total Real time: 24.0 sec
Total Memory Usage: 2331.703125 Mbytes
Reset AAE Options
*** timeDesign #1 [finish] : cpu/real = 0:00:21.4/0:00:23.9 (0.9), totSession cpu/real = 0:00:47.9/0:04:42.5 (0.2), mem = 2331.7M
Innovus 1>

4. Post-Route hold time analysis :



The screenshot shows the iclab018 terminal window with the following content:

```
*** Calculating scaling factor for lib_min libraries using the default operating condition of each library.
Total number of fetched objects 8904
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO: Total number of nets in the design is 8883, 100.0 percent of the nets selected for SI analysis
End delay calculation. (MEM=2349.7 CPU=0:00:02.1 REAL=0:00:02.0)
End delay calculation (fullDC). (MEM=2349.7 CPU=0:00:02.3 REAL=0:00:03.0)
Loading CTE timing window with TwFlowType 0... (CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 2349.7M)
Add other clocks and setupCteToAecClockMapping during iter 1
Loading CTE timing window is completed (CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 2349.7M)
Starting SI iteration 2
Start delay calculation (fullDC) (1 T). (MEM=2322.82)
Glitch Analysis: View av_func_mode_min -- Total Number of Nets Skipped = 0.
Glitch Analysis: View av_func_mode_min -- Total Number of Nets Analyzed = 8904.
Total number of fetched Objects 8904
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO: Total number of nets in the design is 8883, 0.0 percent of the nets selected for SI analysis
End delay calculation. (MEM=2362.99 CPU=0:00:00.0 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=2362.99 CPU=0:00:00.1 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:02.9 real=0:00:03.0 totSessionCpu=0:00:55.2 mem=2363.0M)

-----
timeDesign Summary
-----

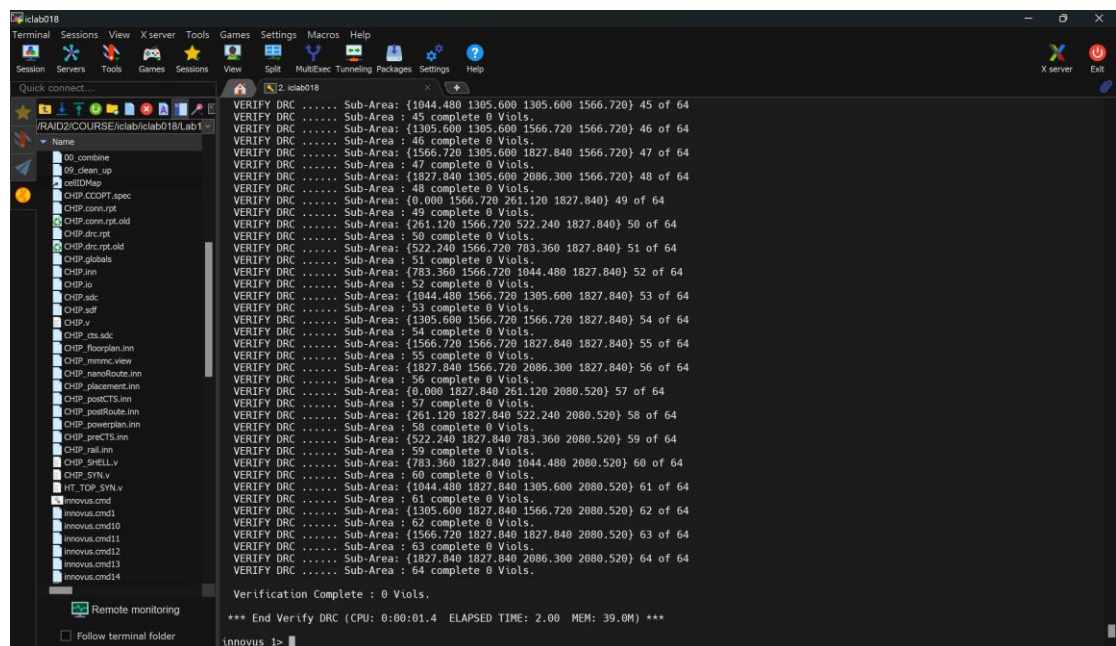
Hold views included:
av_func_mode_min

-----
Hold mode | all | reg2reg | default |
-----|-----|-----|-----|
WNS (ns): | 0.239 | 0.239 | 0.775 |
TNS (ns): | 0.000 | 0.000 | 0.000 |
Violating Paths: | 0 | 0 | 0 |
All Paths: | 371 | 281 | 370 |
-----

Density: 12.682%
(126.814% with Fillers)

-----
Reported timing to dir timingReports
Total CPU time: 3.75 sec
Total Real time: 5.0 sec
Total Memory Usage: 2289.257812 Mbytes
Reset AAE Options
*** timeDesign #2 [finish] : cpu/real = 0:00:03.0/0:00:04.3 (0.9), totSession cpu/real = 0:00:55.4/0:05:50.6 (0.2), mem = 2289.3M
innovus 1>
```

5. DRC result :



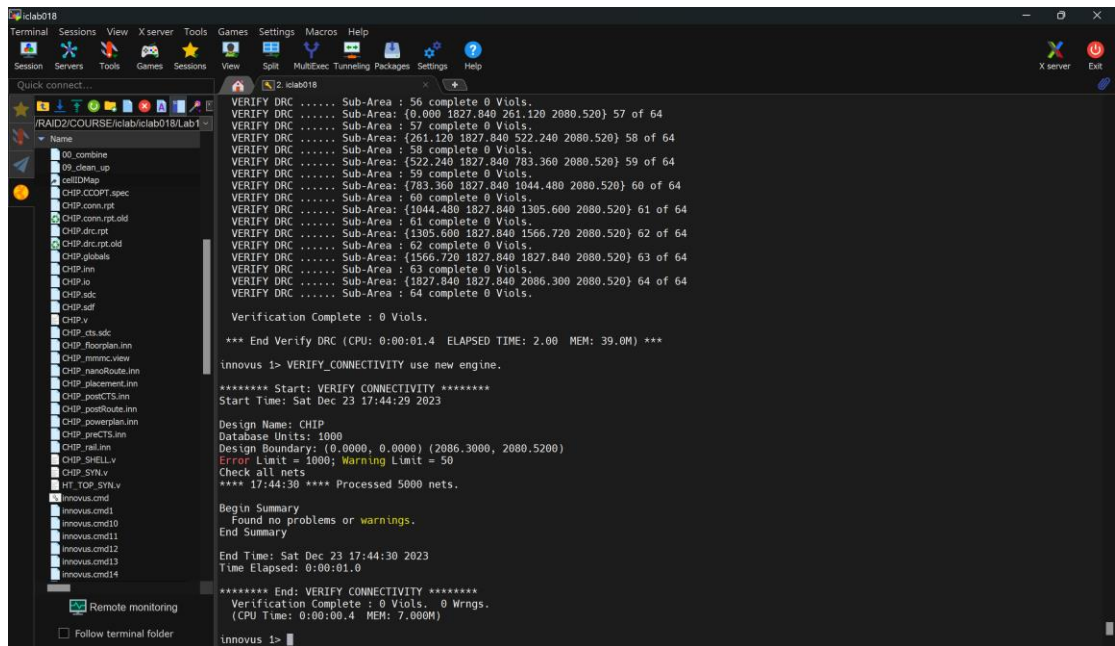
The screenshot shows the iclab018 terminal window with the following content:

```
VERIFY DRC ..... Sub-Area: {1044.480 1305.600 1305.600 1566.720} 45 of 64
VERIFY DRC ..... Sub-Area: 45 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1305.600 1305.600 1566.720 1566.720} 46 of 64
VERIFY DRC ..... Sub-Area: 46 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1566.720 1305.600 1827.840 1566.720} 47 of 64
VERIFY DRC ..... Sub-Area: 47 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1827.840 1305.600 2086.300 1566.720} 48 of 64
VERIFY DRC ..... Sub-Area: 48 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 1566.720 261.120 1827.840} 49 of 64
VERIFY DRC ..... Sub-Area: 49 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {261.120 1566.720 522.240 1827.840} 50 of 64
VERIFY DRC ..... Sub-Area: 50 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {522.240 1566.720 783.360 1827.840} 51 of 64
VERIFY DRC ..... Sub-Area: 51 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {783.360 1566.720 1044.480 1827.840} 52 of 64
VERIFY DRC ..... Sub-Area: 52 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1044.480 1566.720 1305.600 1827.840} 53 of 64
VERIFY DRC ..... Sub-Area: 53 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1305.600 1566.720 1566.720 1827.840} 54 of 64
VERIFY DRC ..... Sub-Area: 54 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1566.720 1566.720 1827.840 1827.840} 55 of 64
VERIFY DRC ..... Sub-Area: 55 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1827.840 1566.720 2086.300 1827.840} 56 of 64
VERIFY DRC ..... Sub-Area: 56 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 1827.840 261.120 2080.520} 57 of 64
VERIFY DRC ..... Sub-Area: 57 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {261.120 1827.840 522.240 2080.520} 58 of 64
VERIFY DRC ..... Sub-Area: 58 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {522.240 1827.840 783.360 2080.520} 59 of 64
VERIFY DRC ..... Sub-Area: 59 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {783.360 1827.840 1044.480 2080.520} 60 of 64
VERIFY DRC ..... Sub-Area: 60 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1044.480 1827.840 1305.600 2080.520} 61 of 64
VERIFY DRC ..... Sub-Area: 61 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1305.600 1827.840 1566.720 2080.520} 62 of 64
VERIFY DRC ..... Sub-Area: 62 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1566.720 1827.840 1827.840 2080.520} 63 of 64
VERIFY DRC ..... Sub-Area: 63 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1827.840 1827.840 2086.300 2080.520} 64 of 64
VERIFY DRC ..... Sub-Area: 64 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:01.4 ELAPSED TIME: 2.00 MEM: 39.0M) ***
innovus 1>
```

6. LVS result :



```
iclab018
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions
Quick connect...
RAID2/COURSE/iclab/iclab018/Lab1
Name
00_combine
00_clean_up
cellIDMap
CHIP.CCOPt.speac
CHIP.constrpt
CHIP.constrpt.old
CHIP.drc.rpt
CHIP.drc.rpt.old
CHIP.globals
CHIP.inn
CHIP.io
CHIP.sdc
CHIP.sdf
CHIP.v
CHIP.vts.sdc
CHIP.floorplan.inn
CHIP.mnmc.view
CHIP.nanoRoute.inn
CHIP.placement.inn
CHIP.postCTS.inn
CHIP.postRoute.inn
CHIP.powerplan.inn
CHIP.preCTS.inn
CHIP.rail.inn
CHIP.SHELL.v
CHIP.SYN.v
HT_TOP_SYN.v
innovus.cnd
innovus.cnd1
innovus.cnd10
innovus.cnd11
innovus.cnd12
innovus.cnd13
innovus.cnd14
Remote monitoring
Follow terminal folder

VERIFY DRC ..... Sub-Area : 56 complete 0 Viols.
VERIFY DRC ..... Sub-Area : {0.000 1827.840 261.120 2080.520} 57 of 64
VERIFY DRC ..... Sub-Area : 57 complete 0 Viols.
VERIFY DRC ..... Sub-Area : {261.120 1827.840 522.240 2080.520} 58 of 64
VERIFY DRC ..... Sub-Area : 58 complete 0 Viols.
VERIFY DRC ..... Sub-Area : {522.240 1827.840 783.360 2080.520} 59 of 64
VERIFY DRC ..... Sub-Area : 59 complete 0 Viols.
VERIFY DRC ..... Sub-Area : {783.360 1827.840 1044.480 2080.520} 60 of 64
VERIFY DRC ..... Sub-Area : 60 complete 0 Viols.
VERIFY DRC ..... Sub-Area : {1044.480 1827.840 1305.600 2080.520} 61 of 64
VERIFY DRC ..... Sub-Area : 61 complete 0 Viols.
VERIFY DRC ..... Sub-Area : {1305.600 1827.840 1566.720 2080.520} 62 of 64
VERIFY DRC ..... Sub-Area : 62 complete 0 Viols.
VERIFY DRC ..... Sub-Area : {1566.720 1827.840 1827.840 2080.520} 63 of 64
VERIFY DRC ..... Sub-Area : 63 complete 0 Viols.
VERIFY DRC ..... Sub-Area : {1827.840 1827.840 2086.300 2080.520} 64 of 64
VERIFY DRC ..... Sub-Area : 64 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:01.4 ELAPSED TIME: 2.00 MEM: 39.0M) ***

innovus 1> VERIFY_CONNECTIVITY use new engine.

***** Start: VERIFY CONNECTIVITY *****
Start Time: Sat Dec 23 17:44:29 2023

Design Name: CHIP
Database Units: 1000
Design Boundary: (0.0000, 0.0000) (2086.3000, 2080.5200)
Error Limit = 1000; Warning Limit = 50
Check all nets.
**** 17:44:30 **** Processed 5000 nets.

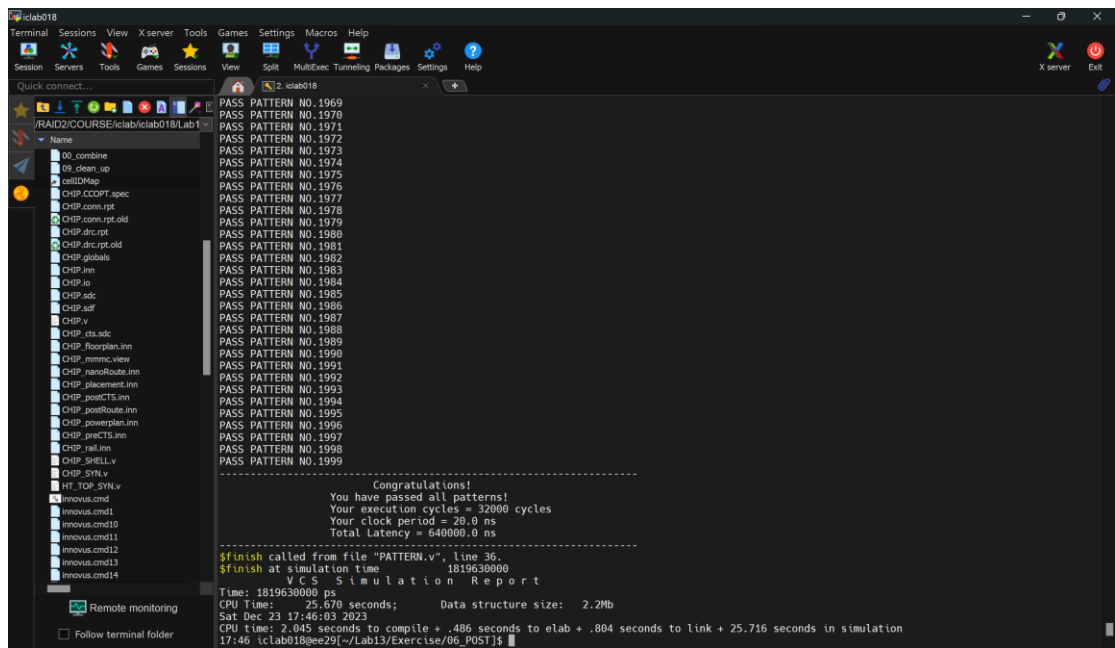
Begin Summary
Found no problems or warnings.
End Summary

End Time: Sat Dec 23 17:44:30 2023
Time Elapsed: 0:00:01.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 0 Viols. 0 Wrngs.
(CPU Time: 0:00:00.4 MEM: 7.000M)

innovus 1>
```

7. Post Layout simulation result :



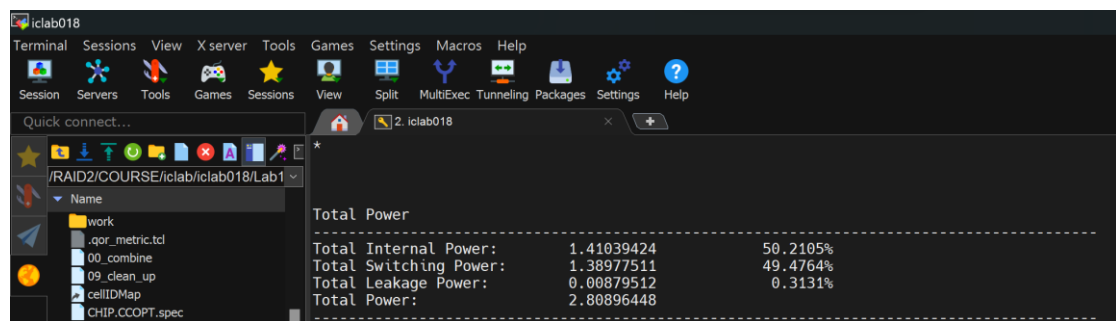
```
iclab018
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions
Quick connect...
RAID2/COURSE/iclab/iclab018/Lab1
Name
00_combine
00_clean_up
cellIDMap
CHIP.CCOPt.speac
CHIP.constrpt
CHIP.constrpt.old
CHIP.drc.rpt
CHIP.drc.rpt.old
CHIP.globals
CHIP.inn
CHIP.io
CHIP.sdc
CHIP.sdf
CHIP.v
CHIP.vts.sdc
CHIP.floorplan.inn
CHIP.mnmc.view
CHIP.nanoRoute.inn
CHIP.placement.inn
CHIP.postCTS.inn
CHIP.postRoute.inn
CHIP.powerplan.inn
CHIP.preCTS.inn
CHIP.rail.inn
CHIP.SHELL.v
CHIP.SYN.v
HT_TOP_SYN.v
innovus.cnd
innovus.cnd1
innovus.cnd10
innovus.cnd11
innovus.cnd12
innovus.cnd13
innovus.cnd14
Remote monitoring
Follow terminal folder

PASS PATTERN NO.1969
PASS PATTERN NO.1970
PASS PATTERN NO.1971
PASS PATTERN NO.1972
PASS PATTERN NO.1973
PASS PATTERN NO.1974
PASS PATTERN NO.1975
PASS PATTERN NO.1976
PASS PATTERN NO.1977
PASS PATTERN NO.1978
PASS PATTERN NO.1979
PASS PATTERN NO.1980
PASS PATTERN NO.1981
PASS PATTERN NO.1982
PASS PATTERN NO.1983
PASS PATTERN NO.1984
PASS PATTERN NO.1985
PASS PATTERN NO.1986
PASS PATTERN NO.1987
PASS PATTERN NO.1988
PASS PATTERN NO.1989
PASS PATTERN NO.1990
PASS PATTERN NO.1991
PASS PATTERN NO.1992
PASS PATTERN NO.1993
PASS PATTERN NO.1994
PASS PATTERN NO.1995
PASS PATTERN NO.1996
PASS PATTERN NO.1997
PASS PATTERN NO.1998
PASS PATTERN NO.1999

-----
Congratulations!
You have passed all patterns!
Your execution cycles = 32000 cycles
Your clock period = 20.0 ns
Total Latency = 640000.0 ns
-----

$finish called from file "PATTERN.v", line 36.
$finish at simulation time 1819630000
VCS Simulation Report
Time: 1819630000 ps
CPU Time: 25.670 seconds; Data structure size: 2.2Mb
Sat Dec 23 17:46:03 2023
CPU time: 2.045 seconds to compile + .486 seconds to elab + .804 seconds to link + 25.716 seconds in simulation
17:46 iclab018@ee29[~/Lab13/Exercise/06_POST]$
```


8. Power result :



9. IR Drop Results :

增加 power ring 與 power stripe 的數量來降低 IR drop 的程度

